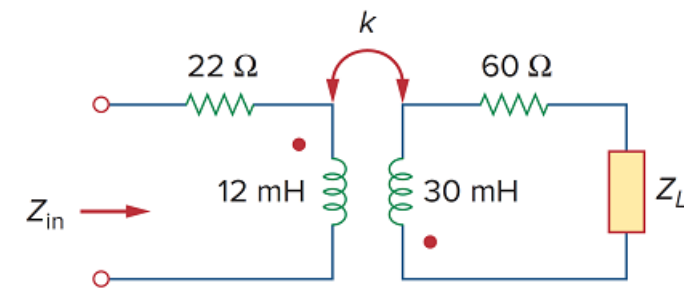


Problem

In the circuit of Fig. 13.86, Z_L is a 15-mH inductor having an impedance of $j40\ \Omega$. Determine Z_{in} when $k = 0.6$.

Figure 13.86



Step-by-step solution

Step 1 of 5

Consider the following circuit diagram:

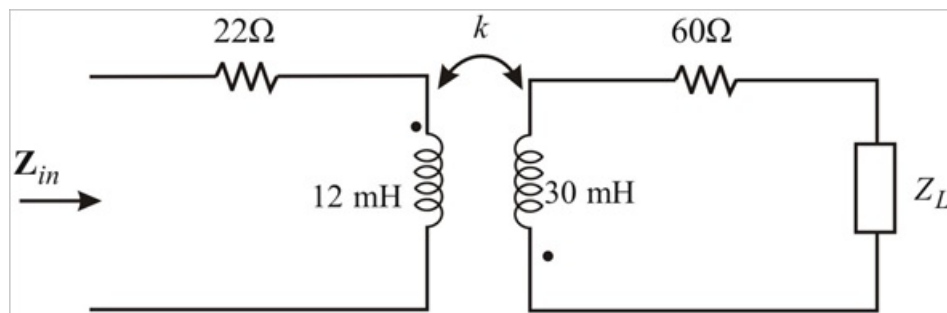


Figure 1

Step 2 of 5

Consider that $Z_L = j40 \Omega$,

And inductance is 15 mH.

$$Z_L = j\omega L$$

$$j40 = j\omega(15 \times 10^{-3})$$

$$\omega = 2666.7 \text{ rad/sec}$$

Coefficient of coupling, $k = 0.6$

Calculate the mutual inductance.

$$\begin{aligned} M &= k\sqrt{L_1 L_2} \\ &= 0.6\sqrt{12 \times 10^{-3} \times 30 \times 10^{-3}} \\ &= 11.384 \text{ mH} \end{aligned}$$

Calculate the inductive impedances.

$$\begin{aligned} Z_{L_1} &= j(2666.7)(12 \times 10^{-3}) \\ &= j32 \Omega \end{aligned}$$

$$\begin{aligned} Z_{L_2} &= j(2666.7)(30 \times 10^{-3}) \\ &= j80 \Omega \end{aligned}$$

$$\begin{aligned} Z_M &= j(2666.7)(11.384 \times 10^{-3}) \\ &= j30.384 \Omega \end{aligned}$$

Step 3 of 5

Convert the circuit in to frequency domain.

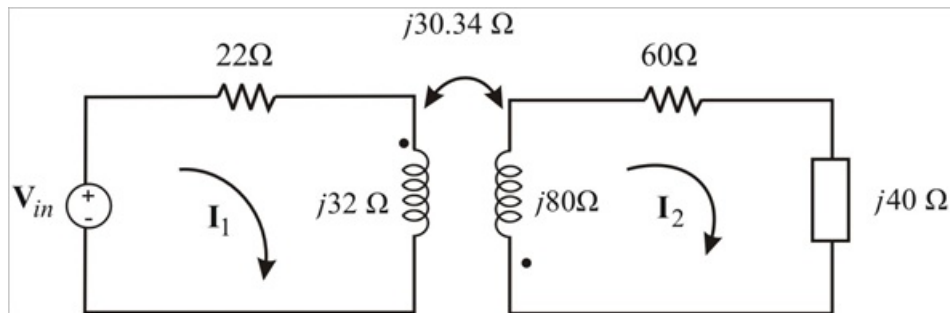


Figure 2

Step 4 of 5

Apply Kirchhoff's voltage law to first loop.

$$V_{in} + 22I_1 + j32I_1 + j30.34I_2 = 0$$

$$I_1(22 + j32) + j30.34I_2 = V_{in}$$

$$I_1(22 + j32) + j30.34I_2 = V_{in} \dots\dots (1)$$

Apply Kirchhoff's voltage law to second loop.

$$(j40 + 60 + j80)I_2 + j30.34I_1 = 0$$

$$I_2(60 + j120) + j30.34I_1 = 0$$

$$I_2 = (-0.202 - j0.101)I_1$$

Step 5 of 5

Substitute this expression for $\mathbf{I}_2$ in equation (1).

$$\mathbf{I}_1 (22 + j32) + j30.34 (-0.202 - j0.101) \mathbf{I}_1 = \mathbf{V}_{in}$$

$$\mathbf{V}_{in} = \mathbf{I}_1 (25.07 + j25.86)$$

$$\frac{\mathbf{V}_{in}}{\mathbf{I}_1} = 25.07 + j25.86$$

$$\mathbf{Z}_{in} = 25.07 + j25.86 \, \Omega$$

Thus, the impedance $\mathbf{Z}_{in}$ is $25.07 + j25.86 \, \Omega$.